

SIDDHANT GAUTAM

Department of Electrical and Computer Engineering
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Professional Summary

Postdoctoral Research Fellow in Electrical and Computer Engineering at the University of Michigan, specializing in computational imaging, inverse problems, and scientific machine learning. Research focuses on developing optimization-driven and physics-informed deep learning frameworks for adaptive sampling and accelerated acquisition in MRI and X-ray computed tomography.

Research Interests

Computational imaging, inverse problems, signal processing, optimization, machine learning, deep learning.

Education

Michigan State University	East Lansing, MI
Ph.D., Computational Mathematics, Science and Engineering	Aug. 2020 – May 2026
• Thesis: <i>“Learning-based Sampling and Reconstruction Methods for Accelerated MRI and X-ray CT”</i>	
• Advisor: Prof. Saiprasad Ravishankar	GPA: 3.78/4.00

Indian Institute of Technology Madras	Chennai, India
M.S., Electrical Engineering	Jul. 2017 – Jun. 2020
• Thesis: <i>“Soil Moisture Retrieval Using Sliced Regression Inversion Technique”</i>	
• Advisor: Prof. Uday K. Khankhoje	GPA: 7.58/10.00

National Institute of Technology Allahabad	Prayagraj, India
B.Tech., Electronics and Communication Engineering	Jul. 2013 – Jun. 2017
• GPA: 8.60/10.00	

Research Experience

University of Michigan	Ann Arbor, MI
Postdoctoral Research Fellow, Electrical and Computer Engineering	Jun. 2026 – Present

Michigan State University	East Lansing, MI
Research Assistant, Computational Mathematics, Science and Engineering	Aug. 2020 – May 2026

Dolby Laboratories	Sunnyvale, CA
Video Processing Research Intern	Jun. 2022 – Aug. 2022

Indian Institute of Technology Madras	Chennai, India
Research and Teaching Assistant, Electrical Engineering	Jul. 2017 – Jun. 2020

Awards and Honors

2026 Ginther Research Fellowship, Department of Computational Mathematics, Science and Engineering, Michigan State University. Competitive departmental fellowship awarded to top graduate students based on prior research achievements, proposed research impact, and faculty support.

- 2025 Outstanding Graduate Student Award**, College of Engineering, Michigan State University. Recognized as the top doctoral student by the faculty committee.
- 2024 Ginther Outstanding Research Award**, Department of Computational Mathematics, Science and Engineering, Michigan State University.
- 2023 Best Student Paper Award Finalist**, Optica Imaging Congress, Boston, MA, USA.
- 2021 Best Paper Award**, CMSE Department Research Symposium, Michigan State University.
- 2020 Best Master’s Thesis Award**, IEEE Geoscience and Remote Sensing Society (GRSS), Kerala Chapter, India.

Publications

Peer-Reviewed Journal Articles

1. **S. Gautam**, A. Li, N. Seiberlich, J. A. Fessler, and S. Ravishankar. “Scan-Adaptive MRI Undersampling Using Neighbor-Based Optimization (SUNO).” *IEEE Transactions on Computational Imaging (TCI)*, 2025. [doi:10.1109/TCI.2026.3653330](https://doi.org/10.1109/TCI.2026.3653330).
2. **S. Gautam**, M. L. Klasky, B. T. Nadiga, T. Wilcox, G. Salazar, and S. Ravishankar. “Learning Robust Features for Scatter Removal and Reconstruction in Dynamic ICF X-Ray Tomography.” *Optics Express*, vol. 33, no. 13, pp. 21741–21758, 2025. [doi:10.1364/OE.565478](https://doi.org/10.1364/OE.565478).

Peer-Reviewed Conference Papers

1. S. Liang, I. R. Alkhouri, **S. Gautam**, Q. Qu, and S. Ravishankar. “UGoDIT: Unsupervised Group Deep Image Prior Via Transferable Weights.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [OpenReview](#).
2. **S. Gautam**, A. Li, and S. Ravishankar. “Patient-Adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering.” *ICASSP 2024 - IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 2081–2085. [doi:10.1109/ICASSP48485.2024.10446528](https://doi.org/10.1109/ICASSP48485.2024.10446528).
3. **S. Gautam**, M. L. Klasky, and S. Ravishankar. “Scatter Removal in Dynamic X-Ray Tomography using Learned Robust Features.” *Computational Optical Sensing and Imaging*, Optica, 2023.
4. A. Choudhary, G. Su, and **S. Gautam**. “Residual Encoding Using Neural Field for Image Sequence Modeling.” *Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2023. [doi:10.1109/IEEECONF59524.2023.10476959](https://doi.org/10.1109/IEEECONF59524.2023.10476959).
5. **S. Gautam**, S. V. Chidambaram, N. Gunturu, and U. K. Khankhoje. “Retrieval of Soil Moisture Using Sliced Regression Inversion Technique.” *Progress in Electromagnetic Research Symposium (PIERS)*, IEEE, 2019. [doi:10.1109/PIERS-Spring46901.2019.9017425](https://doi.org/10.1109/PIERS-Spring46901.2019.9017425).

Preprints and Works Under Review

1. Y. Sun*, **S. Gautam***, C. Huang, N. Seiberlich, I. R. Alkhouri, and S. Ravishankar. “Dynamic MRI Reconstruction Via Dual Deep Priors and Low-Rank Plus Sparse Modeling.” (* equal contribution) arXiv: [arXiv:2605.18709](https://arxiv.org/abs/2605.18709), 2026.
2. **S. Gautam**, A. Li, P. Agarwal, A. K. Attili, J. A. Fessler, N. Seiberlich, and S. Ravishankar. “Scan-Adaptive Dynamic MRI Undersampling Using a Dictionary of Efficiently Learned Patterns.” arXiv: [arXiv:2602.13984](https://arxiv.org/abs/2602.13984), 2026.

3. A. Dhar*, **S. Gautam***, and S. Ravishankar. “Learning Scan-Adaptive MRI Undersampling Patterns with Pre-Optimized Mask Supervision.” Submitted to *ICASSP*, 2026. (* equal contribution) arXiv: [arXiv:2509.16846](https://arxiv.org/abs/2509.16846).

Workshop Papers and Presentations (Non-Archival)

- A. Dhar, **S. Gautam**, and S. Ravishankar. “Pre-Optimized Mask-Guided Scan-Adaptive MRI Undersampling.” *ISMRM Annual Meeting*, 2026.
- **S. Gautam**, A. Li, J. A. Fessler, N. Seiberlich, and S. Ravishankar. “Scan-Adaptive Undersampling for Dynamic Cardiac MRI Using a Collection of Learned Sampling Patterns.” *ISMRM Annual Meeting*, 2026.
- **S. Gautam**, A. Li, N. Seiberlich, J. Fessler, S. Ravishankar. ”Scan-Adaptive MRI Undersampling Using Neighbor-based Optimization.” *IMSI Workshop on Computational Imaging*, 2024.
- **S. Gautam**, A. Li, S. Ravishankar. ”Patient-Adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering.” *Medical Imaging Meets NeurIPS Workshop*, NeurIPS 2023.
- **S. Gautam**, A. Li, S. Ravishankar. ”Patient-Adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering.” *Midwest Machine Learning Symposium (MMLS)*, 2023.

Patents

- **Residual Encoding and Decoding Using a Neural Field**, Inventors: Guan-Ming Su, Anustup Choudhary, **Siddhant Gautam**. Filed by Dolby Laboratories, 2023. (Patent pending).

Invited Talks

- *Patient-Adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering*, Department of EECS, University of Michigan, Ann Arbor, MI, May 2024.
- *Scatter Removal in Dynamic X-Ray Tomography using Learned Robust Features*, Machine Learning for Scientific Imaging, Electronic Imaging Symposium, Burlingame, CA, January 2024.

Teaching Experience

Michigan State University	East Lansing, MI
Teaching Assistant – Data Visualization Principles and Techniques	Spring 2022
Indian Institute of Technology Madras	Chennai, India
Teaching Assistant – Engineering Electromagnetics	Fall 2019
Teaching Assistant – Computational Electromagnetics (NPTEL)	Spring 2019
Teaching Assistant – Electric Circuits and Networks	Fall 2018
Teaching Assistant – Signals and Systems	Spring 2018

Professional Service

Journal Reviewer

IEEE Transactions on Medical Imaging (TMI), *IEEE Transactions on Computational Imaging (TCI)*, *IEEE Journal of Selected Topics in Signal Processing (JSTSP)*.

Conference Reviewer

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), *IEEE International Symposium on Biomedical Imaging (ISBI)*.

Professional Societies

IEEE (Student Member),Optica (Student Member),ISMRM (Student Member).

Technical Skills

Languages: Python, MATLAB, C++

Frameworks: PyTorch, NumPy, SciPy, Matplotlib, Scikit-Learn

Tools: Git, LaTeX